

Integrating Sentiment Analysis into Football Scouting: A Data-Driven Approach

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Abstract—Football scouting has undergone a significant transformation with the integration of data-driven methods. However, existing approaches focus primarily on quantitative performance metrics, while overlooking nuanced insights from qualitative data such as text data. This study explores the potential of sentiment analysis in football scouting by analyzing fan comments on social media platforms to evaluate player perception. A manually labeled dataset comprising 200 comments in four languages (English, German, French, and Spanish) was collected and processed using three sentiment analysis models: VADER, XLM-T, and GPT-4o-mini. A comparative evaluation based on accuracy, precision, recall, and F1-score demonstrated that GPT-4o-mini outperformed the other models, achieving an accuracy of 82% and balanced accuracy of 77%. To translate these insights into practical value, a user-friendly dashboard was developed, enabling scouts to visualize sentiment trends and player-specific feedback through intuitive charts and sample comment displays. The dashboard was presented to a scouting department of a club from the German Bundesliga, where expert feedback confirmed the relevance of sentiment analysis as a supplementary tool for player evaluation and transfer decisions. The results highlight the value of combining quantitative and qualitative insights in modern football scouting, suggesting that sentiment analysis can improve decision-making and strategic planning in player recruitment and market positioning.

Index Terms—Sentiment Analysis, GPT-4o, Football Scouting, Social Media Analysis, Dashboard

I. INTRODUCTION

With an estimated fan base of 3.5 billion people across all continents, football (soccer) is the most popular sport in the world [1]. Beyond its considerable entertainment value, football also exerts a marked social influence. For instance, it has the capacity to foster social cohesion among individuals from diverse backgrounds [2]. Furthermore, football has the potential to serve as a diplomatic instrument, contributing to the enhancement of international relations [3].

Apart from its social impact, football is also a significant economic factor that generates billions of euros every year. Football clubs operate as commercial enterprises and contribute not only to the sports industry, but also to the regional and national economy through sponsorship, broadcasting rights and matchday revenue [4], [5].

In this highly competitive and financially driven environment, one of the most important parts of a football club's operations is player scouting. Identifying talented and suitable players is crucial for sporting success and the associated financial stability [6]. Traditionally, scouting relied primarily

on the subjective judgements of scouts based on personal observations and personal networks. While effective in certain contexts, this approach was resource-intensive, inconsistent, and prone to human bias [7]. In recent decades, scouting methods have been revolutionized by technological advances and the increase in the availability of data. Scientific work emphasizes the integration of machine learning (ML) and the analysis of data to identify suitable players. The focus here is primarily on quantitative data and metrics such as player performance and tactical suitability [8], [9]. For example, advanced statistics such as expected goals (xG) or plus-minus ratings have become standard tools for evaluating on-pitch contributions, allowing clubs to make more informed decisions [10], [11].

However, although data-driven scouting with quantitative data offers clear benefits, Pavitt et al. [12] demonstrate that an exclusive focus on numerical data tends to overlook the nuanced and valuable qualitative insights embedded in textual information. Social media platforms, forums, and fan commentary provide a vast and untapped source of information on public perceptions and attitudes towards players. Sentiment analysis as a branch of natural language processing (NLP), deals with the extraction and classification of opinions, emotions and attitudes from text data, enabling a structured knowledge acquisition [13]. By applying sentiment analysis to textual data, it offers clubs the opportunity to uncover emerging trends, identify hidden potential, and even detect controversies or challenging phases in a player's career. While previous studies have extensively explored sentiment analysis for marketing, political campaigns, and public opinion monitoring [14], [15], its application to football scouting remains underexplored. Integrating sentiment analysis into scouting practices could provide clubs with a scalable and resource-efficient tool to capture collective fan perception and systematically incorporate qualitative insights into player evaluations.

By bridging the gap between quantitative data and qualitative insights, this research contributes to the evolution of football scouting practices, highlighting the value of natural language processing in sports analytics. The key contributions of this research are as follows:

- 1) Application of sentiment analysis to extract and evaluate fan sentiments from football-related social media comments, providing a novel perspective to complement traditional scouting methods.

- 2) Comparative analysis of sentiment analysis models, including lexicon-based approaches and large language models (LLMs), to identify the most effective model for this domain.
- 3) Development and implementation of a user-friendly dashboard, enabling intuitive visualization and practical use by scouting stakeholders.
- 4) Evaluation of the dashboard's usability and relevance through expert feedback, leading to the identification of open issues.

The paper is organized as follows: Section II contains related work in this field. Section III details the methodological approach, including a comparative evaluation of sentiment analysis models as well as the development and integration of a custom dashboard designed to visualize sentiment analysis results. Section IV presents the findings of the model evaluation and its implementation within the dashboard, illustrated through an example analysis. Section V critically examines these findings, discussing their theoretical contributions and practical implications within the context of football scouting. Finally, Section VI concludes the paper by summarizing the main contributions, acknowledging research limitations, and proposing avenues for future investigation.

II. RELATED WORK

In the field of sports, several approaches have been developed to utilise sentiment analysis as a tool to analyze public opinion on matches, players and more general topics. Applications range from assessing fan engagement to predicting outcomes based on social media comments. This section reviews previous work in two categories: broader sports contexts and football-specific applications.

A. Sentiment Analysis in broader sports context

In the broader sports context, sentiment analysis has primarily been used to assess public reactions to sensitive topics in sports. For example, Tirdad et al. [16] analyzed tweets on the topic of concussions during the 2018 FIFA World Cup, demonstrating the effectiveness of neural network-based models to capture public opinion on health issues in sport. Their findings illustrate how real-time sentiment analyzes can influence public discourse and policy decisions in sports medicine. Similarly, Ardianto et al. [17] analyzed public opinion on e-sports education programmes in tweets and compared the application of a Naive Bayes classification with a Support Vector Machine to analyze sentiment. The results of the study suggest that sentiment towards the educational programmes is ambivalent. Furthermore, the authors concluded that the Naive Bayes approach has a higher accuracy in analyzing sentiment. Text mining and sentiment analysis has also been used to predict sports performance in domains other than football. Maymin [18] investigated the use of ML algorithms and text mining techniques to predict transitions from the American college league (NCAA) to the premier professional basketball league (NBA). The study incorporated various data sources, including combined metrics, field statistics and raw

text data from scouting reports, to train a random forest model. The results showed that this approach outperformed the draft choices of most NBA teams and generated an estimated \$100 million in value.

B. Sentiment Analysis in football

In the field of football, various studies have focussed on football-specific sentiment analyzes. In this context, Wunderlich and Memmert [19] investigated the general applicability of lexicon-based sentiment analysis to football-related tweets. The aim of the study was to evaluate the accuracy of these approaches in comparison to manual classification. The study showed that these analysis tools can be an effective method for analyzing sports-related text data. Furthermore, the authors emphasized that sports-specific lexicons should be developed to maximize accuracy. The sentiment lexicon of Aloufi and El Saddik [20], developed for football-related tweets, is a notable example of this approach, facilitating effective sentiment detection for football events. Their work emphasizes the importance of domain-specific lexicons for improving the accuracy of lexicon-based approaches. In addition, Schumaker et al. [21] investigated the role of fan sentiment from Twitter in predicting match outcomes in the Premier League. The aim was to predict the result based on the sentiment in tweets and to use this information for betting decisions. Their model showed a promising return on investment for betting, despite slightly lower accuracy, highlighting the potential of sentiment analysis in real-world applications. A more specific approach in the area of scouting was presented by Kolbinger [22]. He explored the use of text mining and sentiment analysis for various football-related applications, including fan reactions to VAR (Video Assistant Referee) decisions and the integration of scouting reports into prediction models. His work showed how text data can improve performance analysis and decision making in football. Furthermore, he showed how the inclusion of textual information from scouting reports significantly improves the predictive accuracy of models, emphasising the potential of text mining and sentiment analysis for the advancement of performance analysis.

While existing studies in football-specific contexts have explored sentiment analysis for customised lexicons [20], validated lexicon-based tools [19] and investigated the role of social media sentiment in predicting match outcomes [21], this study advances the field by comparing traditional methods and advanced language models for sentiment analysis. Unlike Kolbinger's work [22], which focused on incorporating textual data from scouting reports to improve predictive models, this study prioritizes the use of advanced sentiment analysis techniques. By analyzing social media comments, it aims to leverage collective fan perception to support football scouting. In addition, this work will be differentiated by the development of a user-friendly dashboard that visually displays the findings of the sentiment analysis to support decision making and the evaluation and recruitment of players.

III. METHODOLOGY

The methodology of this paper follows the Design Science Research (DSR) Approach of Hevner et al. [23]. The DSR methodology provides a structured framework for the iterative design, development and evaluation of artefacts to solve practical problems while contributing to theoretical knowledge. In this context, the methodology involves systematically comparing sentiment analysis models and integrating the most effective one into a dashboard, designed to assist scouting departments in extracting and visualizing qualitative insights from social media data. An overview of the Methodology is shown in Fig. 2.

A. Data

This study uses data collected on social media platforms, specifically X and Instagram, and focusses on user comments on professional football clubs and players. A dedicated dataset was systematically created through a multi-step data collection and pre-processing process to ensure a representative sample.

The original data basis includes user comments specifically targeting discussions under official posts of professional football clubs to capture sentiments towards players. Since players from all over the world are observed in scouting, comments from 17 clubs from four major European leagues (with differing national languages) were collected:

- Germany (Bundesliga): 5 clubs
- England (Premier League): 6 clubs
- Spain (La Liga): 3 clubs
- France (League 1): 3 clubs

In total, 85,000 comments were gathered, encompassing a wide range of discussions, including player performances, transfers, and fan sentiment. These data serve as the foundation for all further steps in this study.

To ensure that the dataset was structured and relevant for the player-specific sentiment analysis, several pre-processing and enrichment steps were carried out, which are described below. Once the comments were obtained, a player-specific dataset was created by randomly selecting ten players per club from their squad lists, which were sourced from Transfermarkt.de¹. The number of players was limited to ten for reasons of resource efficiency, ensuring that the dataset remained manageable while still providing a representative sample. The comments were then filtered for the occurrence of at least one part of the player's name and matched to the relevant players to ensure that the extracted data focussed on discussions about individual players. The resulting dataset was structured by linking each comment to the respective club and player.

To further refine the dataset for model evaluation, a manually labeled subset was created. A total of 200 comments were selected and divided into the four main leagues and languages: 50 German, 50 English, 50 French and 50 Spanish comments. Each comment was manually labeled with one of three sentiment categories: POSITIVE, NEGATIVE or NEUTRAL. The distribution of sentiments (Fig. 1) is 143

POSITIVE, 29 NEGATIVE and 28 NEUTRAL. This labeled subset serves as a benchmark for evaluating the performance of different sentiment analysis models for multiple languages.

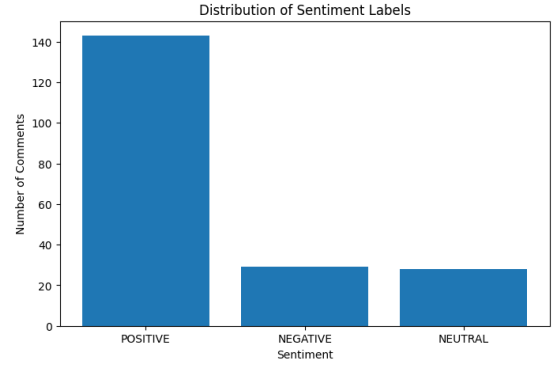


Fig. 1. Sentiment Label distribution in validation dataset.

B. Model Comparison

To evaluate the effectiveness of different models for sentiment analysis in football scouting, this study compares three different models: VADER, XLM-T, and GPT-4o-mini. Each model represents a distinct approach to sentiment classification, ranging from lexicon-based methods to transformer-based deep learning architectures.

VADER: VADER (Valence Aware Dictionary for sEntiment Reasoning) [24] is a simple rule and lexicon based model developed for sentiment analysis of social media posts. When analyzing the sentiment of so-called microblog content, challenges often arise due to the short text length or the use of abbreviations or emoticons. By combining a human-generated lexicon with heuristic rules, VADER presents an approach to solve these challenges.

Since VADER is optimized for the English language, in this study, all comments in German, French, and Spanish were first translated into English using the python library “translate” to maintain consistency across the dataset. After translation, VADER calculated polarity scores for each comment, generating four key sentiment metrics: positive, negative, neutral, and a compound score. The compound score, which aggregates the intensity of sentiments within the comment, was used to assign a sentiment label. The classification was based on the following thresholds:

- Positive: Compound score ≥ 0.05
- Neutral: Compound score between -0.05 and 0.05
- Negative: Compound score ≤ -0.05

XLM-T: XLM-T (Cross-lingual Language Model for Twitter) [25] is a transformer-based language model, which builds up on the architecture of XLM-R (Cross-lingual Language Model-RoBERTa) [26], aiming to enhance multilingual sentiment analysis, particularly for social media contexts such as twitter. While XLM-R was pretrained on large-scale multilingual datasets from Common Crawl covering over 100 languages, XLM-T focuses on the informal and dynamic nature

¹www.transfermarkt.de

of social media data. This includes dealing with challenges like slang, abbreviations or emojis. A major advance of XLM-T over XLM-R is the integration of adapter modules that enable more efficient fine-tuning of the model. Instead of re-training the entire model, only small adapter layers are updated, reducing computational costs and improving performance. Despite this efficiency, XLM-T retains the strong cross-language transfer capabilities of XLM-R and achieves superior performance on multilingual sentiment classification tasks [25], [26].

In this study, all comments from the validation dataset were processed using XLM-T. For each comment, the model generated a predicted sentiment label (positive, negative or neutral) along with a sentiment score between -1 and 1.

GPT 4o-mini: GPT-4o-mini is a transformer-based language model based on the autoregressive architecture of the Generative Pre-trained Transformer (GPT) family. In contrast to encoder-based models such as XLM-R, which are optimised for classification tasks through masked language modelling, GPT-4o-mini uses a pure decoder architecture. This enables the model to generate coherent, context-rich responses based on sequential input data, making it highly effective for natural language generation and understanding tasks. One of GPT-4o-mini's key strengths is its ability to capture complex linguistic structures. This generative ability allows moods to be interpreted not just on the basis of isolated keywords, but taking into account the broader context of discourse [27], [28].

All comments from the validation dataset were processed using GPT-4o-mini. Unlike the other models, GPT-4o-mini was provided with two inputs: the comment from a social media platform as well as the name of the footballer to be investigated regarding the sentiment against him. This approach allowed GPT-4o-mini to specifically evaluate the sentiment towards the given player, ensuring that the sentiment analysis was not influenced by irrelevant parts of the comment. To guide the model's output, a custom prompt was used:

"You are a linguistics expert and have been assigned to rate social media comments from soccer fans. In the following, you will be given the name of a soccer player and a comment from a social media channel. Please rate it according to the sentiment towards the player. The score should be in a range from -1 to 1, where -1 is negative, 0 is neutral, and 1 is positive. Please also consider whether the comment really refers to the specified player and which part of the comment refers specifically to the player. Also note the emojis. Please provide a sentiment score (ranging from -1 to 1), a sentiment label (NEGATIVE, NEUTRAL, or POSITIVE), and an explanation."

For each comment-player pair GPT-4o-mini generated: A sentiment score ranging from -1 (negative) to 1 (positive), a corresponding sentiment label (NEGATIVE, POSITIVE or NEUTRAL) as well as an explanation detailing how the sentiment was determined.

C. Dashboard Development

The dashboard was developed as the central artifact within the DSR framework to allow the scouting department to integrate qualitative fan sentiment analysis into the player evaluation process. Its goal is to complement the data-driven scouting process with structured insights from fan opinions.

System Architecture: The dashboard was implemented using Python and relies on three core technologies. The frontend was developed with Streamlit to provide an intuitive and interactive web interface. For data handling, transformation, and aggregation, the dashboard uses the Python libraries Pandas and Numpy. Finally, Plotly is used to generate dynamic charts and graphs that visualize the sentiment analysis results. This architecture ensures that the dashboard can efficiently handle large datasets and provide actionable insights, supporting the scouting department in making informed decisions based on fan sentiment.

Input Modes: The dashboard supports two distinct input modes to balance flexibility and efficiency: (1) classification and visualization of new, unseen data and (2) re-evaluation of already processed data without requiring new computations. This dual approach optimises performance and costs by minimising redundant calculations while providing the ability to process new social media data on demand.

(1) In the first mode, users submit comments in JSON format, along with player and club information. The model processes the data, generates sentiment scores and labels, and stores the results in a structured DataFrame, which forms the basis for visualization. This mode enables the processing and analysis of fan opinions, providing scouts with structured insights into player perception. Since processing through a sentiment analysis model can be time-intensive, the dashboard includes a download function that allows the processed data to be exported as a CSV file. This enables users to retain the processed data for future reference and eliminates the need to reprocess the same dataset multiple times, thereby improving efficiency.

(2) The second mode is designed for the re-evaluation of previously processed data. Users can upload a CSV file containing the pre-processed sentiment data. The dashboard directly visualizes the data without re-running the sentiment analysis, which significantly reduces computational time and costs, particularly for large datasets.

Visualization: The dashboard includes several visualization components designed to provide comprehensive insights into fan sentiment and enable scouts to identify patterns and trends effectively. The average sentiment score is visualized using a gauge chart, which provides an overview of the general sentiment towards each player. The score ranges from -1 (negative) to +1 (positive), allowing scouts to quickly identify whether fan perception is predominantly positive or negative. The sentiment score distribution is displayed as a histogram, showing how sentiment scores are spread across the dataset. This helps identify whether fan opinions are polarized or balanced.

A word cloud highlights the most frequently mentioned terms in fan comments, with the font size reflecting the frequency of occurrence. This allows scouts to quickly identify recurring themes and key topics associated with a player. To improve clarity, stop words are excluded from the word cloud. Sample Comments are shown for each sentiment category (positive, neutral, and negative). Initially, only one representative comment per sentiment category is displayed to provide a quick overview. By clicking on a sample comment, the dashboard expands to display all other comments classified with the same sentiment. The expanded view includes the sentiment score and (if available) the explanation for each comment, enabling scouts to explore detailed sentiment patterns and polarising topics.

All in all, the dashboard was designed to support flexible sentiment analysis. The combination of quantitative visualizations (e.g., sentiment scores and distribution) with qualitative insights (e.g., word cloud and comment samples) provides a comprehensive understanding of fan perception. The practical application of the dashboard is demonstrated and discussed in Section IV and V.

IV. RESULTS

This section presents the results of the model comparison, evaluates the performance of each sentiment analysis model, and demonstrates the practical application of the developed dashboard.

A. Model Comparison Results

The comparison and evaluation focuses on standard performance metrics such as accuracy, balanced accuracy, precision, recall, and F1-score [29], [30]. The evaluation was performed on a validation dataset of 200 manually labeled comments.

As shown in Table I GPT-4o-mini achieved the highest accuracy of 0.82 and balanced accuracy of 0.77. The model is outperforming XLM-T (0.65, 0.58) and VADER (0.38, 0.44).

TABLE I
PERFORMANCE METRICS OF SENTIMENT ANALYSIS MODELS

Model	Acc.	Bal. Acc.	Prec.	Rec.	F1
GPT-4o-mini	0.82	0.77	0.71	0.77	0.74
XLM-T	0.65	0.58	0.52	0.58	0.54
VADER	0.38	0.44	0.43	0.44	0.35

Table II highlights the performance of the models in the different classes. VADER performed poorly, especially for the NEUTRAL and NEGATIVE classes, where the recall was below 0.35. The model favored predicting the POSITIVE class, leading to imbalanced results. XLM-T significantly outperformed VADER, achieving an overall accuracy of 65%. This model demonstrated a better balance across all classes, particularly with higher recall for the NEGATIVE class. The performance for the NEUTRAL class remains weak, with low precision (0.29) and recall (0.43). The best and robust performance across all classes was achieved by GPT-4o-mini. It showed a very good precision and recall for the

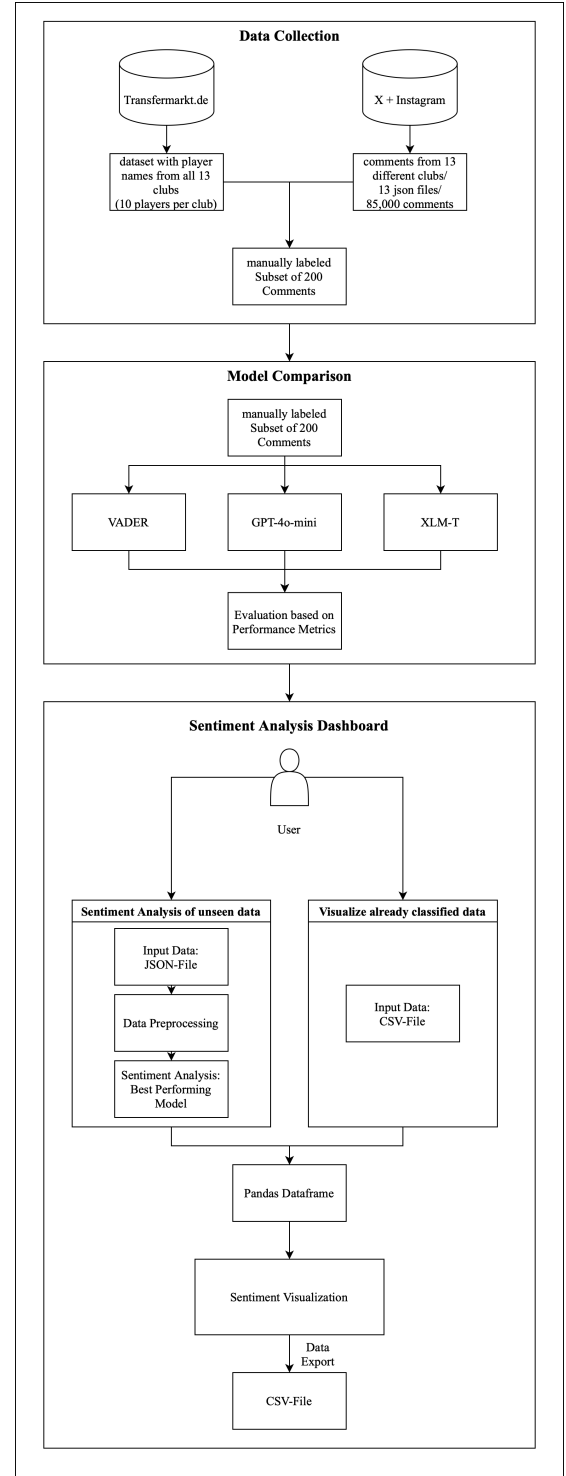


Fig. 2. Visualization of the Methodology

POSITIVE and NEGATIVE classes. While performance for the NEUTRAL class remains weaker, it is notably better than the other models. While XLM-T struggles with the NEUTRAL class and VADER is heavily biased towards the POSITIVE class, GPT-4o-mini is used for this application due to its high accuracy and consistency.

TABLE II
CLASSIFICATION REPORT PER SENTIMENT CLASS

Model	Label	Prec.	Rec.	F1
GPT-4o-mini	POS	0.94	0.86	0.90
	NEU	0.49	0.64	0.55
	NEG	0.72	0.79	0.75
XLM-T	POS	0.86	0.70	0.77
	NEU	0.29	0.43	0.35
	NEG	0.42	0.62	0.50
VADER	POS	0.87	0.34	0.48
	NEU	0.18	0.64	0.28
	NEG	0.23	0.34	0.28

B. Dashboard Results

Based on the results of the model comparison, GPT-4o-mini was selected as the sentiment analysis model integrated into the dashboard due to its superior classification performance. To evaluate the practical application of the dashboard, an example analysis was conducted focusing on Cole Palmer, as seen in Figure 3. Palmer was selected due to his recent outstanding performances and growing public attention. The goal of the analysis was to assess the sentiment of the fans towards the player and how this reflects his performance on the pitch.

A total of 1117 comments related to Cole Palmer were collected from social media platforms. The comments originated between August and October 2024. The data was processed using GPT-4o-mini, which completed the analysis in 2027 seconds (approximately 34 minutes), resulting in an average processing time of 1.815 seconds per comment. The model generated a sentiment score between -1 (negative) and +1 (positive), a sentiment label (positive, neutral or negative), and an explanation for the classification.

The sentiment analysis revealed an average sentiment score of 0.507, indicating an overall positive fan perception of Palmer. 66.6% of the comments related to Cole Palmer were positive, 19.7% were neutral, and 13.7% were negative. The increase in positive sentiment can be linked to Palmer’s exceptional form during the analyzed period, where he accumulated a significant number of goal contributions and achieved a historic four-goal performance in a single match. This trend reflects a strong alignment between on-field performance and fan perception.

Sample comments are shown for each sentiment category (positive, neutral, and negative). An example for a positive comment is “What if City have Palmer and KDB on the mid, I think Haaland will get 70+ goals” with a sentiment score of 1.0. The comment shows a clear positive sentiment and highlights the potential Manchester City could have if Cole Palmer had stayed and was now playing alongside Kevin De Bruyne and Erling Haaland. A negative comment like “Palmer is bot” was classified with a sentiment score of -1.0. The explanation highlighted the term “bot” as a derogatory expression, driving the negative classification.

The word cloud highlighted terms such as “cold”, “ice cold”, “goal” or “assist” as frequently associated with Palmer whereas negative ones are not. These frequently used words indicate his high scoring threat and ice-cold behaviour in front

of the goal and show that his performance on the pitch is reflected in the prevailing language of the fans.

The sentiment analysis of Cole Palmer revealed a strong correlation between his on-field success and positive fan perception. The dashboard effectively captured this relationship by visualizing sentiment patterns and highlighting key moments. The combination of sentiment scores, sentiment distribution, and explanatory insights provided a comprehensive understanding of how Palmer’s exceptional goal contributions influenced fan sentiment.

The example demonstrated that sentiment analysis using GPT-4o-mini provides valuable insights into fan perception. The consistency between positive sentiment and Palmer’s on-field performance suggests that fan sentiment could serve as an additional metric in player evaluation. The dashboard’s ability to visualize sentiment patterns and provide explanatory feedback can improve the scouting department’s understanding of fan opinion.

V. DISCUSSION

This section discusses the model comparison and the applicability of the application in the scouting area in which it was evaluated by experts. The results of this study were presented to the scouting department of the german Bundesliga club VfB Stuttgart 1893 e.V., where the dashboard was demonstrated as a tool for integrating sentiment analysis into the scouting process. Following the demonstration, an expert interview was conducted with members of the scouting team to gather detailed feedback on the relevance, usability, and potential improvements of the dashboard. The insights gained from this feedback informed the development of the practical implications and provided valuable guidance for future enhancements.

A. Interpretation of Model Comparison

The model comparison demonstrated that GPT-4o-mini outperformed both VADER and XLM-T across all performance metrics, confirming its suitability as the sentiment analysis model integrated into the dashboard. GPT-4o-mini achieved the highest overall accuracy (0.82) and balanced accuracy (0.77), reflecting its ability to provide consistent and reliable classification across different sentiment categories. Its superior performance underscores the strength of transformer-based models, particularly in handling complex language structures and contextual sentiment. A key strength of GPT-4o-mini was its high precision and recall for the POSITIVE (0.94, 0.86) and NEGATIVE (0.72, 0.79) classes, indicating that the model was not only able to identify positive and negative sentiment accurately but also to maintain consistency when faced with ambiguous or mixed-language input. While the performance for the NEUTRAL class was relatively lower (precision = 0.49, recall = 0.64), GPT-4o-mini still outperformed the other models in this category, reflecting its enhanced capacity to detect subtle sentiment expressions.

What sets GPT-4o-mini apart from the other models is its ability to process contextual sentiment and to recognise the parts of the comment that relate precisely to the desired

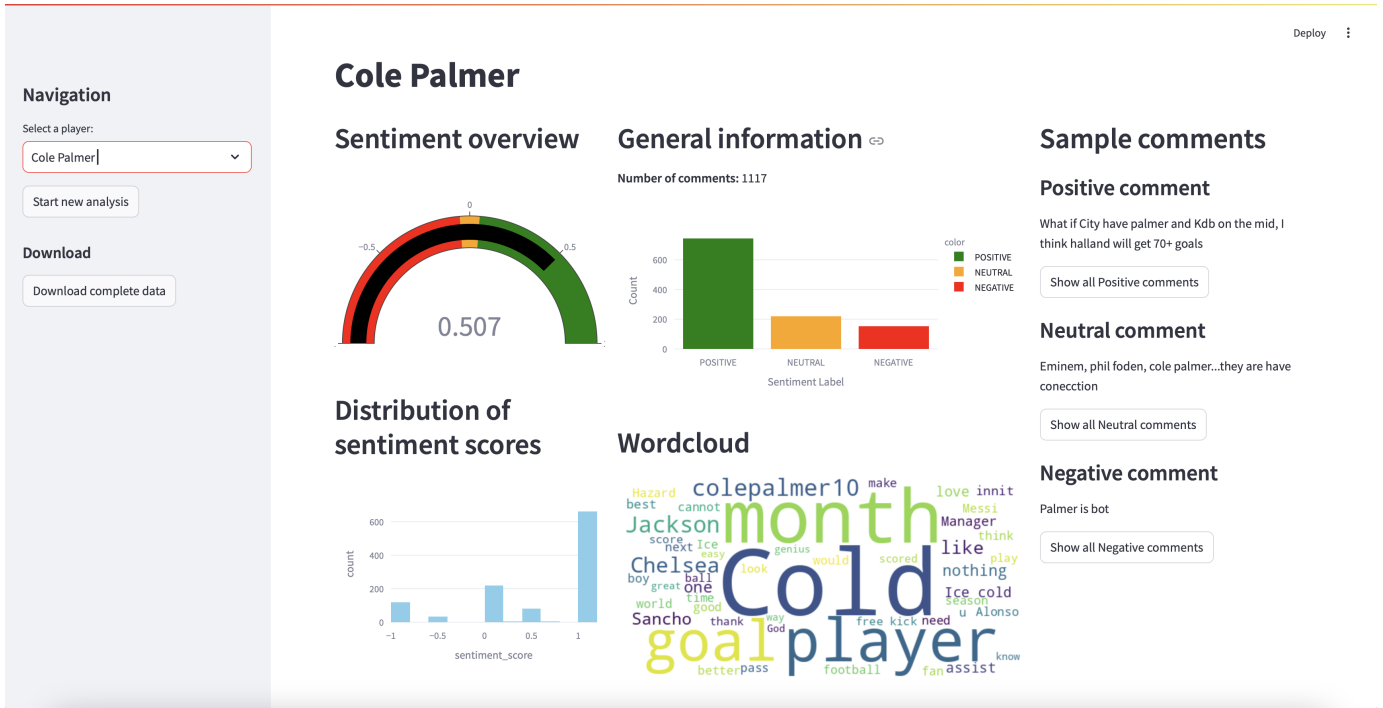


Fig. 3. Screenshot of the classification from the example analysis.

player. This is possible because the model was provided with both the comment and the name of the player as input, allowing it to focus on sentiment directed toward the specified player rather than general statements or off-topic remarks. By linking the player name to the comment, GPT-4o-mini could better identify the sentiment context and disregard irrelevant or misleading language. Unlike VADER and XLM-T, which rely primarily on keyword-based or direct sentiment cues, GPT-4o-mini interpreted the underlying sentiment based on the broader linguistic structure of the comments and the given input. This was particularly valuable in football-related social media data, where fan language is often emotionally charged and context-dependent. Sarcasm, implicit criticism, and mixed sentiment were handled more accurately by GPT-4o-mini due to its decoder-only transformer architecture and autoregressive language processing capabilities.

In contrast, VADER and XLM-T showed weaker performance due to their inherent limitations in handling complex and context-dependent language. VADER, as a lexicon-based model, assigns fixed sentiment values to words without considering the surrounding context, leading to misclassifications in cases of sarcasm or implicit criticism. XLM-T, while moderately better than VADER, struggled with ambiguous language and lacked player-specific context due to its encoder-based architecture.

The superior performance of GPT-4o-mini compared to VADER and XLM-T aligns with findings from previous research. Chinnalagu [31] demonstrated that fine-tuned LLMs consistently outperform both lexicon-based and deep-learning models due to their ability to understand complex language

structures and contextual dependencies. Similarly, Lossio-Ventura et al. [32] highlighted that LLMs, including GPT-based models, achieve higher accuracy and F1-scores than traditional transformer-based models like BERT and lexicon-based models due to their superior handling of mixed-language input and implicit sentiment.

Moreover, GPT-4o-mini’s ability to generate detailed explanations for each classification enhanced its interpretability. The model not only produced a sentiment score and label but also explained why the comment was classified in a particular way, highlighting specific phrases or language patterns that influenced the outcome. This transparency increased trust in the model’s predictions.

The decision to integrate GPT-4o-mini into the dashboard was based on its superior accuracy, balanced performance across sentiment categories, and ability to generate detailed explanations. Its capacity to process mixed and multilingual input, combined with contextual understanding, positioned it as the most effective model for capturing and analyzing fan sentiment. While VADER performed well for simple sentiment classification and XLM-T showed moderate improvements, GPT-4o-mini provided a comprehensive solution by combining high classification performance with enhanced interpretability, making it particularly suited for the nuanced and emotionally charged language of football-related social media data.

B. Feedback from Expert Interviews

To validate the practical applicability of the dashboard, expert feedback was obtained from the scouting department of VfB Stuttgart 1893 e.V.. The interviews provided valuable

insights into the relevance, usability, and improvement potential of the dashboard and the underlying sentiment analysis model.

Relevance and Value for Scouting: The scouts generally viewed the integration of fan sentiment into the scouting process as valuable but emphasized that it would primarily serve as a supplementary tool rather than a core scouting method. Sentiment analysis was considered particularly useful in the final stages of the scouting process when the technical and tactical evaluation of a player is already well established. The perception of the fans was seen as a valuable factor in transfer decisions and contract negotiations, especially when particular anomalies, such as discrepancies between performance data and market interest, occur. An example of this would be if a player's performance data and video analysis were convincing, but the player remained undervalued on the transfer market. A sentiment analysis could reveal negative sentiments about the player's character, which would explain the low interest from the market.

Usability and Design of the Dashboard: The dashboard's intuitive design and interactive visualizations were positively received from the scouts. However, several suggestions for improving user experience were identified. The current presentation of comments was viewed as unclear. Introducing a sorting mechanism (e.g., by sentiment score or time) would enhance usability. While the sentiment distribution was viewed positively, a simpler gauge-like indicator of the average sentiment score was suggested to improve interpretability. The word cloud was perceived very positively, only the use of consistent color coding for each sentiment category would improve clarity. The explanations provided by GPT-4o-mini were considered helpful but were not deemed critical unless the classification appeared incorrect or inconsistent with other scouting data.

Integration in the scouting workflow: The experts emphasised that sentiment analysis via the dashboard has a high chance of being used in the club's scouting process, particularly in the final stages of player evaluation as already mentioned. The combination of sentiment analysis with traditional performance metrics was viewed as a promising strategy for creating a more comprehensive player profile. By correlating fan sentiment with on-field performance, clubs can gain deeper insights into a player's psychological resilience and ability to handle public pressure. They also pointed out that it could be an interesting approach to analyze larger amounts of data from different clubs and locations and then compare the most suitable players from the performance data with the players with the most positive sentiment to find the best match. The experts also suggested that expanding the dashboard's functionality to enable the visualization of sentiment trends over time and across platforms would further enhance its value. Tracking changes in fan perception over time could provide additional insights into how specific events, such as injuries, transfers, or key performances impact public perception. This improvement would allow clubs to respond more effectively to shifts in fan sentiment.

C. Practical Applications

The expert feedback obtained during the interviews directly influenced the identification of several practical applications for sentiment analysis using the built application. The ability to systematically analyze fan sentiment offers new dimensions of player evaluation and market assessment.

(1) Sentiment analysis can be used as a supplementary tool in transfer decisions. Positive fan sentiment towards a player can increase a club's willingness to invest in that player, not only based on sporting potential but also due to the commercial opportunities associated with fan engagement. Conversely, negative sentiment or reputational risks identified through fan comments can serve as a warning signal, prompting further investigation before finalizing a transfer.

(2) Sentiment analysis can provide insights that are valuable in contract negotiations and salary evaluations. Clubs can use fan opinions and social media data analysis to generate information that helps them convince the player to make a transfer. They can also use sentiment analysis to assess the potential impact of a player's departure on fan engagement and merchandise sales.

(3) Sentiment analysis can support market positioning and fan engagement strategies. Players with consistently positive sentiment could be more likely to attract commercial partnerships and increase club visibility, as shown by similar approaches such as Popp et al. [33], who identified a connection between positive social media sentiment and increased season ticket sales.

(4) Beyond its immediate use in player scouting, the sentiment analysis framework could be expanded to other areas within professional football. One promising avenue is the use of sentiment analysis to assess the marketability and commercial potential of players. By analyzing fan sentiment toward individual players, clubs can identify key drivers of fan engagement and optimize their merchandising and sponsorship strategies. For example, understanding which players generate the most positive sentiment could inform decisions about shirt sales, promotional campaigns, and social media engagement strategies. Comparable ideas have been proposed by Yu [34]. Furthermore, sentiment analysis could be extended to evaluate broader club dynamics, such as the perception of managers, coaches, and club leadership. Tracking fan sentiment toward tactical decisions, managerial performance, and transfer activity could provide clubs with valuable feedback. This information could also be used to adjust communication strategies and improve fan relations.

(5) Another potential application is in the area of sports agency and player representation. Player agents could use sentiment analysis to evaluate the public perception of their clients and adjust contract negotiation strategies accordingly. High fan sentiment toward a player could strengthen the agent's position in contract talks, while negative sentiment might prompt early career adjustments or reputation management efforts. Similarly, agencies could use sentiment analysis to identify rising talents with strong fan support, guiding

recruitment decisions and portfolio management.

The flexibility and scalability of the sentiment analysis application make it a powerful tool for both strategic and operational decision-making within professional football. The ability to extend sentiment analysis beyond player evaluation and integrate it into broader club management, commercial strategy, and competitive analysis highlights its potential as a versatile asset in modern football operations.

VI. CONCLUSION

This study developed a user-friendly dashboard that enables the analysis of fan sentiment toward football players based on social media data. The dashboard processes comments using GPT-4o-mini, which was identified as the most effective sentiment analysis model through a comparative evaluation with VADER and XLM-T. GPT-4o-mini achieved the highest overall accuracy (0.82) and balanced accuracy (0.77), demonstrating its ability to handle complex, context-dependent language more effectively than the other models. This research extends existing work on sentiment analysis in football-specific contexts in several key areas. While previous studies have focused on customized lexicons [20], validated lexicon-based tools [19], and the role of social media sentiment in predicting match outcomes [21], this study advances the field by systematically comparing traditional sentiment analysis methods with advanced LLMs. The results demonstrate that transformer-based models, particularly GPT-4o-mini, offer superior performance in handling complex, multilingual, and context-dependent sentiment expressions. Furthermore, unlike previous work [22], which integrated textual data from scouting reports to improve predictive models, this study focused on analyzing social media comments to leverage collective fan perception. This approach provides qualitative insights into player perception, which complements traditional performance metrics used in scouting. The developed dashboard was presented to a Bundesliga scouting department from VfB Stuttgart 1893 e.V., where it was positively received. The scouts confirmed that sentiment analysis provides valuable complementary insights to traditional scouting methods. The ability to assess fan perception was viewed as a strategic advantage, particularly in the final stage of the scouting process. The feedback underscores the potential of sentiment analysis to enhance decision-making in professional football and support a more comprehensive player evaluation framework.

A. Limitations & Future Work

Several limitations of this study should be acknowledged, which provide opportunities for future improvement. The validation dataset consisted of only 200 manually labeled comments (50 per league), which limits the generalizability of the results. The dataset was imbalanced, with a higher proportion of positive comments, which may have affected the overall evaluation metrics. Expanding the validation dataset and balancing the distribution of sentiment classes could increase the robustness of the model evaluation and improve consistency across sentiment categories. Additionally,

the comments lacked timestamps, preventing any analysis of sentiment evolution over time. Incorporating timestamps would enable sentiment analysis over time, providing deeper insights into how fan perception evolves in response to specific events such as matches [35], injuries, or transfers. While GPT-4o-mini demonstrated strong performance in identifying positive and negative sentiment, the classification of neutral sentiment remained weaker. This suggests that further fine-tuning or an alternative model architecture may be needed to improve the model's ability to distinguish neutral sentiment from mixed or ambiguous input. Providing GPT-4o-mini with additional context [36] could improve the model's ability to distinguish between neutral and mixed sentiment. Further refinement through prompt engineering [37] (for example, including examples of ambiguous and neutral comments) could also improve the model's interpretability and accuracy. The data sources were limited to Instagram and X, which may have introduced platform-specific biases. Expanding the data collection to other platforms, such as Reddit, TikTok, and football-specific forums, would improve the diversity of sentiment data and increase the generalizability of the model's outputs. Different platforms exhibit distinct communication styles and engagement patterns, which could provide more comprehensive insights into fan sentiment and player perception. Another limitation relates to the filtering of comments during pre-processing. The current filtering approach is limited to the player's first and last name, which may exclude relevant data due to variations in spelling or nickname usage. Introducing models to catch more name variations [38] in the pre-processing phase could improve the completeness of the data by capturing a wider range of relevant comments and increasing the overall robustness of the sentiment analysis. The dashboard currently operates only in batch mode and does not support real-time sentiment processing, limiting the system's ability to track sentiment changes during live events or over extended periods. Implementing real-time processing [39] would enable sentiment analysis during matches or transfer windows, allowing clubs to respond more quickly to fan sentiment changes. Certain limitations also exist in the visualizations. For example, the Word Cloud does not differentiate between sentiment categories through color coding, which reduces the clarity of sentiment distribution. Improving this by assigning distinct colors to different sentiment categories could make the sentiment patterns more immediately recognizable. Additionally, expanding the dashboard's filtering options to include criteria such as data source, sentiment score, and time frame would enhance its flexibility and allow for more targeted and detailed analysis. Finally, future research could explore the integration of sentiment data with player performance metrics. This would involve comparing the most suitable players identified through sentiment analysis with those highlighted by performance metrics. Developing a composite evaluation framework that aligns qualitative fan perception with quantitative performance data could provide scouting departments with a more holistic approach to identifying the best candidates for specific squad positions.

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